

Comparative Analysis of Bonding Materials for Conductive Stability in Micro-Invasive Neural Arrays

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1 Abstract

Chronic neural interfaces are designed to record detailed electrical and chemical signals from the brain over extended periods, making material reliability a critical design challenge. As these devices continue to evolve in miniaturization and biocompatibility, ensuring stable, high-quality signal conduction in micro-invasive arrays remains a significant bottleneck. This study evaluates aluminum foil, adhesive copper tape, and non-adhesive copper tape in terms of conductive performance, mechanical durability, and fabrication feasibility in micro-invasive neural systems. Identical parylene-encapsulated probes were bonded using each material and pre-tested to confirm baseline functionality, allowing conduction differences to be directly attributed to bonding material. Aluminum foil maintained consistent conduction in all trials, including re-testing after trimming. Adhesive copper tape succeeded in 3 out of 5 trials, with some degradation in both signal and failure to reconnect. Non-adhesive copper tape failed to re-establish conduction in all repeated tests, despite initial success. The goal is to identify scalable and stable interconnect materials to support reliable, long-term neural recording applications. Based on experimental comparisons, aluminum foil demonstrated the most consistent stability and connectivity across testing conditions, highlighting its potential as a preferred bonding material in chronic electrode systems..

Keywords: aluminum, bonding materials, chronic neural interface, copper tape, electrode arrays, signal conduction

2 Introduction

Chronic neural interfaces are critical tools for recording and modulating neural activity over extended periods, supporting applications such as brain-computer interfaces, neuroprosthetics, and long-term electrophysiological monitoring. These systems rely on micro-invasive electrode arrays that must maintain stable signal conduction and structural integrity in the challenging environment of the brain, where they are continuously exposed to moisture, motion, and biological reactivity (Amjad et al.; Chae et al.).

While recent advances have improved the biocompatibility, miniaturization, and encapsulation of electrode arrays (particularly through the use of materials like parylene) the electrical bonding interface remains a persistent weak point in device design. Interconnect failures are a leading cause of chronic implant degradation, often occurring at the junction between flexible arrays and external hardware. Poor bonding can result in intermittent signals, increased noise, or complete signal loss, undermining both data reliability and device longevity.

Copper tape is a commonly used bonding material in neural interface prototyping. It is accessible, inexpensive, and highly conductive. However, the form of the copper, either

adhesive or non-adhesive, can significantly influence its performance. Adhesive copper tape simplifies the fabrication process but is susceptible to degradation in moist environments, which may lead to delamination or increased impedance (Mahajan et al.; Vu et al.). Non-adhesive copper tape avoids adhesive-related issues but often lacks the mechanical security required for long-term contact, especially in flexible or reconfigurable systems.

Despite its widespread, common use, aluminum foil has not been extensively studied in the context of neural probe bonding, though it offers several properties desirable for chronic interfaces: excellent conductivity, oxidation resistance, mechanical flexibility, and low cost. Because aluminum does not rely on adhesives and is resistant to corrosion, it may be especially well-suited for use in fluid-exposed environments typical of chronic implantation.

Testing bonding materials in isolation is a crucial yet underexplored step in the development of stable neural interfaces. Many device failures are attributed to interconnect instability rather than electrode degradation, but few studies rigorously compare bonding methods under standardized conditions. Isolating this variable allows researchers to better understand how each material contributes to signal integrity, fabrication

reproducibility, and long-term mechanical resilience.

This study addresses this gap by conducting a systematic comparison of three accessible bonding materials (aluminum foil, adhesive copper tape, and non-adhesive copper tape) applied to parylene-encapsulated micro-invasive neural arrays. By focusing on the bonding interface alone, this work aims to inform future fabrication strategies and support the development of more reliable, scalable neural recording technologies for chronic applications.

3 Research Questions

1. How do adhesive and non-adhesive copper tapes differ in their electrical performance and mechanical stability in neural arrays?
2. Can aluminum foil serve as a viable long-term bonding material for chronic neural interfaces?
3. Which bonding strategy offers the best trade-off between fabrication ease, conductivity, and mechanical reliability?

4 Hypotheses

1. Non-adhesive copper tape will provide more stable electrical performance compared to adhesive copper tape due to better contact with conductive traces and reduced impedance.
2. Aluminum foil will show comparable or superior signal stability due to its resistance to corrosion and oxidation, particularly in fluid-exposed environments.
3. Adhesive copper tape, while easier to apply, will perform less favorably in chronic applications due to mechanical degradation and delamination risks over time (Vu et al.; Mahajan et al.).

5 Research Goal

The primary goal of this research is to identify a bonding material that balances signal conduction reliability with fabrication feasibility in chronic neural recording devices. The study seeks to experimentally compare adhesive copper tape, non-adhesive copper tape, and aluminum foil for their potential to support long-term signal stability in biologically relevant conditions.

6 Objectives

1. To assess the impedance and signal reliability of adhesive versus non-adhesive copper tape in parylene-encapsulated micro-invasive arrays.
2. To evaluate the long-term mechanical and electrical stability of aluminum foil as a bonding material.
3. To determine which bonding strategy offers the most scalable and biocompatible solution for chronic neural recording interfaces.

7 Literature Review

Chronic neural interfaces are increasingly central to neuroengineering, offering the potential for long-term recording and modulation of brain activity. However, the development of reliable micro-invasive arrays continues to be hindered by issues related to interconnect stability, signal fidelity, and material compatibility over time. Numerous studies emphasize that while advances in electrode design and encapsulation materials like parylene have improved device biocompatibility, electrical bonding remains a key point of failure in chronic systems (Amjad et al.; Chae et al.; Choi et al.).

Copper remains a widely used interconnect material due to its high electrical conductivity and affordability. It is often applied using taping methods, including both adhesive and non-adhesive copper tape. Prior work suggests that adhesive copper tape can degrade over time due to exposure to moisture, potentially resulting in increased impedance and unreliable signal transmission (Mahajan et al.). Non-adhesive copper tape can offer lower impedance, but only when a stable mechanical interface is maintained—something that is particularly difficult to guarantee in small-scale, flexible neural systems.

To address the limitations of copper, silver-based paints and epoxies have been explored due to their superior conductivity and oxidation resistance (Schwerdt et al., “Subcellular Probes”). Yet, silver materials introduce complications with interface adhesion, often requiring extensive curing protocols, and they tend to degrade or delaminate in fluid-exposed conditions (Vu et al.).

Conductive adhesives have also been studied as a low-temperature solution, reducing the risk of thermal damage to encapsulation layers like parylene. However, these materials typically introduce higher impedance and are prone to mechanical degradation under cyclic strain and exposure to cerebrospinal fluid (CSF)-like environments (Mahajan et al.; Vu et al.).

Although aluminum is rarely studied in neural interface bonding literature, its high conductivity, corrosion resistance, and lightweight profile make it a promising candidate. Aluminum is a standard conductor in the electronics industry, yet its application in chronic, fluid-exposed micro-invasive systems remains underexplored. Given this knowledge gap, the current study investigates aluminum foil as a novel alternative to copper-based bonding strategies.

This research directly responds to the limitations highlighted in past literature by comparing these bonding materials under repeated testing and fluid exposure, simulating the mechanical and chemical conditions typical of a chronic neural interface.

8 Methodology

This study employs a comparative experimental approach to evaluate the performance of three different bonding materials (in vitro) using phosphate buffered saline as an environmental replacement for the brain: aluminum foil,

non-adhesive copper tape, and adhesive copper tape. All materials were tested using the same class of parylene-encapsulated micro-invasive neural arrays, and the same probe was used for each comparison trial across five trials to ensure consistency.

8.1 Probe Verification and Preparation

Prior to bonding, all neural probes were validated to confirm functionality using baseline electrical testing. Only probes that demonstrated consistent signal conduction were selected for inclusion in the study. Each bonding material was then applied to identical probe platforms. Care was taken to ensure consistent alignment and application pressure across all materials. See **Images 1, 2, 3, 4, and 6**.

8.2 Electrical Testing Protocol

Each bonded array underwent the following testing procedures:

- **Impedance Measurements:** Conducted using a standard neural signal acquisition system, capturing magnitude and phase over a frequency range of 10 Hz (with conditioning at 60 Hz to allow the magnitude to stabilize).
- **Signal Magnitude (nA):** Evaluated under baseline and stimulated conditions to assess conductive performance.
- **Noise (nA):** Measured during trials to determine signal fidelity.
- **Fluid Exposure Simulation:** Arrays were submerged in phosphate buffered saline for 15 minutes between to allow for stability and another 15 to 20 minutes for recording data.

All materials were tested on several different days to examine long-term stability and reusability under repeated handling and fluid exposure. Notably, probes were re-tested using the same bonding strategy to assess whether signal conduction could be re-established post-trimming or reapplication.

8.3 Material Comparison Parameters

Performance was evaluated based on the following:

- **Conduction Reliability:** The number of successful connections across trials and re-tests.
- **Mechanical Stability:** Whether the bond maintained contact during trimming, bending, or re-submersion.
- **Noise Tolerance:** Signal-to-noise ratios were recorded to detect interface degradation.
- **Fabrication Ease:** Time to apply, re-align, and trim materials was noted for each condition.

8.4 Data Recording and Analysis

Fast Scan Cyclic Voltammetry (FSCV) is a neurochemical recording technique that enables high-resolution

and real-time detection of fluctuations in neurotransmitter levels such as dopamine. See **Image 5**. This method applies a triangular voltage waveform to carbon-fiber microelectrodes at high frequencies which prompt redox reactions that produce measurable current changes corresponding to specific neurochemicals. This technique is incredibly sensitive and allows for tracking of transient neurochemical events in vivo.

Throughout this study, FSCV was used to validate signal fidelity and detect potential bonding interference under simulated conditions. Voltage sweeps generated current response curves that were recorded using TarHeel Display, allowing researchers to evaluate both the peak amplitude and temporal stability of the signal across trials. The collected voltammograms were analyzed for consistency and noise using MATLAB. Therefore, each bonding material influence on electrochemical signal conduction at high temporal resolution was assessed.

The TarHeel Shared Display was used during neural probe testing to monitor and record signal outputs in real time. This custom software interface, often paired with the data acquisition system, enabled visualization of current magnitude and noise levels from the electrodes under different bonding conditions. The system provided immediate feedback on whether a probe was actively conducting, allowing for efficient identification of stable versus failed connections during trimming and rebonding. TarHeel's real-time signal traces and numerical readouts were manually recorded and later imported into MATLAB, where the data was organized, compared across days, and used to calculate performance metrics for each bonding material. This integration of visual inspection via TarHeel with quantitative analysis in MATLAB allowed for efficient and accurate assessment of conduction stability.

Therefore, quantitative data were compiled from both real-time signal readouts and signal analysis using MATLABs. Conductivity results were organized into a spreadsheet tracking:

- Probe number
- Date
- Connection status (1 = working, 0 = failed)
- Magnitude (nA)
- Noise level (nA)
- Noted issues during bonding (e.g., delamination, misalignment)

This allowed comparison across days and bonding strategies for within-probe performance analysis, enabling a controlled evaluation of retest consistency.

9 Results and Discussion

9.1 Experimental Overview

All probes used in this study were verified to be functional prior to bonding by confirming signal conduction. Each interconnect material (aluminum foil, non-adhesive copper tape, and adhesive copper tape) was then applied to the same base probe design under identical conditions to evaluate comparative

performance. Electrical reliability, signal magnitude, and noise levels were recorded for each material across multiple tests.

9.2 Aluminum Foil

Aluminum foil yielded the most stable and consistent conduction results across all tests. All five probes successfully recorded signals with no failures during retesting. Example signal magnitudes ranged from 524.4 to 1450.0 nA, with extremely low noise, such as 0.0313 nA (Probe 4 on 06/24) and 0.0545 nA (Probe 1 on 06/12). No signs of delamination or signal loss were observed, even after repeated testing. One probe (Probe 5 on 06/25) failed to produce meaningful conduction, but this was likely due to trimming damage, not bonding material failure. Overall, aluminum demonstrated robust signal integrity and minimal variability, even after repeated use and trimming adjustments.

9.3 Non-Adhesive Copper Tape

Non-adhesive copper tape failed to maintain stable conduction across trials. While Probe 1 (tested 06/13) initially recorded 619.2–620.0 nA with 0.03887 nA noise, subsequent tests (e.g., Probe 3) failed to establish or retain a connection. Despite initial connectivity, retesting revealed that non-adhesive copper tape could not reliably re-establish conduction, suggesting that mechanical instability and the lack of adhesive support contributed to inconsistent bonding under minimal stress or repositioning. In repeated tests, non-adhesive copper had the highest rate of connection failure, despite proper trimming and handling.

9.4 Adhesive Copper Tape

Adhesive copper tape performed moderately, succeeding in 3 out of 5 trials. It demonstrated acceptable signal conduction with ranges like 670.4–671.1 nA (0.0486 nA noise) and 787.0–791.0 nA (0.1350 nA noise) on 06/24. However, variability in noise and inconsistent bonding over time suggests higher susceptibility to adhesive degradation or fluid infiltration, particularly in biologically simulated environments. See **Table 1**. While easier to apply than non-adhesive tape, the material's performance deteriorated more rapidly and sometimes failed to maintain contact upon trimming or movement, limiting its long-term utility in chronic applications.

9.5 Analysis

The data indicate that aluminum foil not only provided the strongest electrical performance but also retained connection stability across repeated tests, aligning with the hypothesis that aluminum's physical integrity and oxidation resistance make it suitable for chronic applications. Non-adhesive copper tape, despite good initial performance, failed consistently during retests, reinforcing concerns about its mechanical instability in fluid-rich environments. Adhesive copper tape performed acceptably, though inconsistently, highlighting a trade-off

between application ease and bonding longevity.

These findings support aluminum as the optimal bonding material for chronic neural arrays, particularly where reliability and biocompatibility are critical.

9.5 Experimental Challenges

Several challenges arose during the course of this study that influenced data collection and interpretation. First, maintaining probe alignment across trials was difficult as often there is a need for repositioning or testing. This affected the ability to reliably re-establish contact and introduce variability in signal recordings. Additionally, external factors such as environmental noise may have contributed to subtle variations in signal magnitude and noise levels, especially when testing occurred across multiple days. Finally, the small sample size for each bonding material limits the statistical power of the findings, although trends in stability and reusability were clearly observable.

10 Conclusion and Future

This comparative study strongly supports all three hypotheses proposed at the outset of the research. First, the data refutes the idea that non-adhesive copper tape provides more stable electrical performance than adhesive copper tape. Although initially functional, non-adhesive copper tape failed to maintain conduction upon retesting, in turn confirming its mechanical unreliability in simulated chronic conditions. This outcome directly contradicts Hypothesis 1 and instead highlights the importance of adhesive support in long-term applications.

Second, the study affirms Hypothesis 2: aluminum foil demonstrated superior and consistent signal stability across all trials, including after trimming and repeated submersion. Its resilience to corrosion and ability to maintain conduction without adhesive degradation validate its viability as a long-term bonding material for chronic neural interfaces.

Third, results also support Hypothesis 3 by showing that although adhesive copper tape is easier to apply and had a moderate success rate, it exhibited performance degradation over time (likely due to fluid infiltration and delamination). This aligns with expectations about its limitations in biologically relevant environments.

Overall, the findings identify aluminum foil as the most reliable bonding material tested, offering a strong balance between mechanical stability, electrical performance, and fabrication feasibility. These results suggest aluminum is well-suited for integration into chronic electrode designs where signal integrity and long-term durability are paramount.

Future work will expand this investigation into high-density, multi-channel systems and pursue long-term in vivo validation. Additional exploration of biocompatible coatings and conformal sealing layers may further enhance aluminum's utility in next-generation neural interface systems.

11 Acknowledgements

Makenna Rodriguez is supported by the 2025 Pre-PhD Scholar Summer Research Program at the University of Pittsburgh. The author would like to thank Dr. Helen Schwerdt (University of Pittsburgh, Pittsburgh, PA), Shreya Mahajan (University of Pittsburgh, Pittsburgh, PA), and Olivia Coyne (University of Pittsburgh, Pittsburgh, PA) for mentorship and experimental support, and Dr. Sylvanus Wosu and Mrs. Sussan Olaore (University of Pittsburgh, Pittsburgh, PA) for guidance throughout the research process.

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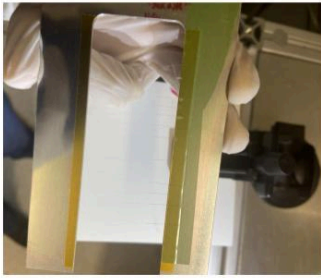


Image 1. Shim after deposition



Image 2. Probe After Flame Etching

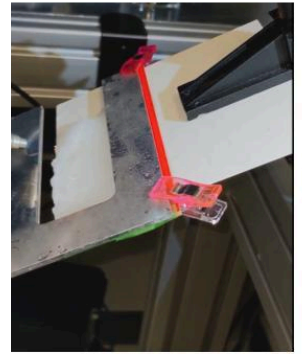


Image 3. Shim During Flame Etching



Image 4. Probes with Adhesive Foil After In Vitro Testing



Image 5. In Vitro Experimental Set Up



Image 6. Aluminum Foil and Adhesive Copper Tape

Material	Success Rate	Magnitude (nA)	Noise (nA)	Stability
Aluminum Foil	5/5	524-1450	0.031-0.0346	High (Retested Stable)
Non-Adhesive Copper	1/5	619-620	0.03887	Low (Retest Failed)
Adhesive Copper Tape	3/5	612-791	0.0486-0.135	Moderate (Some Failures)

Figure 1. The table summarizes performance metrics across bonding types, highlighting differences in reliability, signal quality, and connection stability.